PROJECT DIRECTIVE

CONCURRENT

A modern SameTime conquest game

This is the document I would hand myself before writing a line of code.
It locks the soul of Risk II SameTime, kills the 2000-era friction,
and sequences the build so the engine is real before the map is pretty.

Do not start with Unity. Do not start with a globe renderer.
Start with a deterministic rules kernel and tests that fail loudly.

LOCKS

STATUS	Pre-prototype · rules locked for MVP
CODENAME	CONCURRENT
STACK	TypeScript engine → React/SVG client → multiplayer later
FIRST SHIP	Hotseat 3–5p on a 42-territory map, 60–90 min
NON-GOAL	A licensed Risk clone. Do not use Hasbro IP.

00 — HOW TO USE THIS DOCUMENT

Treat this as standing orders, not inspiration. If a later idea fights a lock in section 01, the lock wins until you explicitly amend the bible and write why. The first playable artifact is not a pretty map. It is a rules engine that can take secret orders from N players and emit a deterministic next world-state.

Work order is fixed: kernel → tests → CLI replay → hotseat UI → map polish → netcode. Skipping ahead is how this becomes another unfinished Risk clone with a globe shader.

01 — NON-NEGOTIABLES

These are the soul. If you cut one, you are making a different game.

Sacred mechanics

- **Simultaneous planning.** Every player writes reinforcements, attacks, and the tactical move before any of them resolve.
- **Committed attackers vacate.** Armies assigned to an attack leave their origin this round and cannot defend it. This is the load-bearing rule. Never soften it.
- **Convergence.** Multiple forces may legally strike the same territory in one resolution pass.
- **Fixed resolution order.** Border clashes → mass invasions → single invasions → spoils → surge. No ad-hoc "whoever clicked first."
- **Restricted reinforcement.** You cannot dump an entire income stack onto one isolated border. Placement is capped by local connectivity.
- **Incomplete information at commit time.** You plan against a last-known map, not against revealed enemy orders.

Hard bans for MVP

- No Hasbro / Risk trademarks, card art, general names, or official map labels used as product identity. Original map names. Original title.
- No real-time unit micromanagement. This is not an RTS wearing a board-game coat.
- No sequential "take your whole turn while others watch" mode in the primary ruleset. Classic mode is a later optional module, not the spine.
- No 5-color custom d12 lookup tables. That system is historically interesting and operationally bad.
- No 4-hour default game length. Target 60–90 minutes at 4 players once people know the rules.

02 — PRODUCT SHAPE

Digital-first board game, not a 4X sim. Hotseat and online async/simultaneous both matter. Physical print-and-play comes after the digital ruleset is proven, because the engine will tell you which numbers actually work.

Layer	What it is	When
L0 Kernel	Pure TS rules engine. No React. No canvas. Deterministic.	Week 1
L1 Harness	CLI + JSON fixtures + golden replays.	Week 1–2
L2 Client	React hotseat. SVG map. Order sheet. Resolution cinema.	Week 2–4
L3 Net	Lockstep room: submit → lock → resolve → broadcast.	After L2 is fun
L4 Content	2nd map, factions, scenarios, solo bot.	After people replay L2

Do not build L3 until a stranger can finish a hotseat game without you in the room.

Recommended stack

- **Language:** TypeScript, strict. Shared types between engine and client.
- **Engine package:** `packages/engine` — zero DOM imports.
- **Client:** Vite + React + SVG. The map is a graph, not a 3D globe.
- **State:** Engine owns truth. UI is a view + order composer.
- **Tests:** Vitest. Combat and resolution order get table-driven tests first.
- **Persistence:** JSON match logs from day one. You will need them to debug "why did Ukraine fall."
- **Later net:** Simple room server (Node) that only stores sealed orders and calls `resolveRound(state, orders[])`.

03 — RULES LOCK FOR MVP

This is the playable contract. Numbers are starting values, not scripture — but the structure is. Tune after ten recorded games, not after one feeling.

3.1 Players, map, victory

- Players: 3–5 for MVP (4 is the design center).
- Map: 42 territories, 6 regions (continents). Use an original geography so you are not shipping a Risk facsimile. Same graph density as classic Risk is fine.
- Setup: deal territories randomly, then simultaneous secret placement of remaining starting armies under the restriction cap.
- Victory: control 38 territories at the end of a full round, *or* be the last player with territories. Secondary table option: highest score after 8 rounds (territories + region bonuses).
- Elimination: a player with 0 territories is out. Their pending attacks still resolve this round if already committed.

3.2 Round phases

Phase	Public?	What happens
0 Diplomacy	Yes	Talk. Optional public declarations. No mechanical lock in MVP.
1 Income	Yes	Engine computes income. Players see their own number.
2 Reinforce	Secret	Place new armies under connectivity cap. Cash one card set optional.
3 Orders	Secret	Declare 0–N attacks and at most one surge flag per successful capture path.
4 Resolve	Public	Engine runs the five-step battle queue. Cinema / log.
5 Tactical	Secret then reveal	One regroup: any number of armies, one origin → one adjacent friendly.
6 Upkeep	Yes	Cards, elimination, victory check, next round.

3.3 Income

Income = max(3, floor((territoriesOwned + largestConnectedBloc) / 3)) + regionBonus + cardSetValue.

Region bonuses start at: small isolated region 2, mid 3–4, large 5–6. Recalibrate so the richest region is tempting but leaky — never a free win. Largest connected bloc is the biggest set of your territories joined by adjacency. This rewards coherent empires without ignoring scattered holdings.

Placement cap: armies you may add to territory T this phase ≤ 1 + (number of T's adjacent territories you already own). Unused income is lost. Do not bank. Loss creates pressure; banking recreates the classic stack-and-wait problem.

3.4 Attack declaration

- An attack is a tuple: {from, to, strength, surge?: boolean}.
- Legal if from is yours, to is adjacent and not yours, strength ≥ 1, and origin retains at least 1 army after all outgoing attacks from that origin are summed.
- All armies in *strength* leave origin immediately when orders lock. They are in transit: they do not defend origin and they do not defend destination until the fight is won.
- You may launch multiple attacks from one origin, and attacks from many origins onto one destination.
- Surge: if the first fight is won, leftover transit armies may immediately contest one pre-declared second territory adjacent to the captured one. Surge happens last in the queue so it cannot pre-empt someone else's primary invasion.

3.5 Combat math (the replacement for colored d12s)

Goal: readable odds, attacker-leaning (SameTime bias), mass still matters, no face-lookup tables. MVP combat is a single opposed roll plus size dice.

Procedure for a pairwise fight (one attacker packet vs current defender stack):

- Attacker dice = 1 + floor((strength - 1) / 3), cap 5.
- Defender dice = 1 + floor((garrison - 1) / 4), cap 4. Defense gets fewer dice but holds ground.
- Each die is a d6. Both sides keep their highest die. Attacker adds +1 if strength ≥ 2× garrison. Defender adds +1 if a fortification marker is present (content later; off in MVP).
- Compare highest totals. Strictly higher wins that volley.
- **Tie: attacker wins the volley.** This is the SameTime thumb on the scale. Keep it.
- Loser removes 2 armies, winner removes 1. If the loser would go to 0, they go to 0 and the fight ends.
- Repeat volleys until one side is gone or the attacker packet is wiped.
- If attacker packet survives and defender is 0, destination flips. Transit survivors occupy it (minimum 1 must stay if a surge is also declared and legal).

Why this and not fully deterministic: SameTime needs some swing so planning is not just spreadsheet dominance. Why this and not the d12 rainbow: a stranger can compute "8 vs 5 is favored but not free" in their head.

3.6 Multi-force resolution

Step A — Border clashes. If A declared into B and B declared into A, resolve those two packets against each other first in the field. Survivors, if any, continue as invasions into the opposing territory with remaining strength. Empty origins stay empty.

Step B — Mass invasions. All remaining packets targeting the same destination fight the defender as a sequence of pairwise volleys in randomized-but-seeded order (seed = hash(roundId + destId)). After the defender is eliminated, leftover attacker packets fight each other for the spoils using the same combat procedure (Step D).

Step C — Single invasions. Everything that is just one packet vs one garrison.

Step D — Spoils. Two or more surviving allied-by-circumstance packets on a vacant or newly emptied destination. They are not allies. They fight until one packet remains.

Step E — Surge. Only packets that (1) flagged surge, (2) captured their primary, (3) have ≥2 armies left. They attack the pre-declared second target. If that target was already taken this round by someone else, the surge hits the new owner.

Seeded RNG is mandatory. Same orders + same seed = same outcome. This is what makes replays, tests, and later netcode possible.

3.7 Cards (keep light)

- On capturing ≥1 territory in a round, draw 1 card at upkeep. Hand limit 5; at 5 you must cash before drawing.
- Sets: three matching types or one of each. Values: 4, then 6, 8, 10, then +5 each additional set cashed *at the table* (global escalation, classic pressure).
- No wild special powers in MVP. Cards are income spikes, not a second game.

3.8 Tactical move

After battles, each player may move any number of armies from exactly one friendly territory to one adjacent friendly territory, leaving ≥1 behind if the origin should remain owned. Planned secretly, revealed together. No chains. This is logistics, not a second attack phase.

04 — DATA MODEL

Put this in `packages/engine/src/types.ts` *first. Names below are the contract.*

```
export type PlayerId = string
export type TerrId = string
export type RoundId = number

export interface Territory {
  id: TerrId
  name: string
  region: string
  adjacent: TerrId[]
  owner: PlayerId | null
  armies: number
}

export interface Player {
  id: PlayerId
  name: string
  color: string
  cards: Card[]
  alive: boolean
}

export interface AttackOrder {
  from: TerrId
  to: TerrId
  strength: number
  surgeTo?: TerrId
}

export interface PlayerOrders {
  playerId: PlayerId
  placements: Record<TerrId, number>
  cashSet?: Card[]
  attacks: AttackOrder[]
  tactical?: { from: TerrId; to: TerrId; strength: number }
}

export interface World {
  mapId: string
  round: RoundId
```

```
  seed: string
  territories: Record<TerrId, Territory>
  players: Record<PlayerId, Player>
  setsCashed: number
  log: Event[]
}
```

Engine public API is three functions and nothing else for a while: `createMatch`, `legalOrders(world, playerId)`, `resolveRound(world, orders[])`. If UI wants convenience helpers, put them in a thin facade — do not leak React into the kernel.

05 — REPO LAYOUT

```
concurrent/
  packages/engine/          # pure TS, vitest
    src/
      types.ts
      map.ts                # graph helpers, connected components
      income.ts
      legal.ts              # order validation
      combat.ts             # volley math
      resolve.ts            # five-step queue
      rng.ts                # mulberry32 from seed
      log.ts
    test/
      combat.test.ts
      resolve-order.test.ts
      income.test.ts
      fixtures/
  packages/maps/
    src/world42.ts          # original 42-node graph + SVG paths
  apps/web/                 # Vite React hotseat
    src/
      ui/Map.tsx
      ui/OrderSheet.tsx
      ui/ResolutionLog.tsx
      state/match.ts
  apps/cli/
    replay.ts               # node concurrent replay match.json
  docs/
    BUILD_BIBLE.pdf         # this file
  README.md
```

06 — FIRST 48 HOURS

If you only follow one section, follow this one. Do it in order.

Hour 0–2 · Scaffold

- pnpm init + workspaces. packages/engine with vitest. Strict TS.
- Commit the types file. Empty resolveRound that throws “not implemented.”
- Add mulberry32 (or equivalent) in rng.ts. No Math.random in the engine. Ever.

Hour 2–6 · Tiny map + income

- Do not build 42 territories yet. Build a 7-territory “knife fight” map: two regions, obvious chokepoint.
- Implement connected-component size and income(). Unit tests for 3 territories vs 3+3 connected.
- Implement placement cap validator.

Hour 6–14 · Combat

• Table-driven tests: 1v1, 8v5, 20v6, 3v12. Assert winner frequency over 1000 seeded rolls sits in a recorded band, plus exact outcomes for fixed seeds.

• Implement pairwise volleys until annihilation.

• Do not tune dice yet. Record the bands. Tuning is week 3 work.

Hour 14–28 · Resolution queue

• Fixture: two players attack each other (border clash).

• Fixture: three packets hit one defender (mass + spoils).

• Fixture: attack + surge into a third territory.

• Fixture: overcommit origin, third player walks into the empty origin. This is the soul test. If this is not dramatic and correct, stop and fix it.

• Every fixture writes a human-readable event log.

Hour 28–40 · CLI match

• apps/cli can start a 3p knife-fight match, accept orders as JSON, print the map as text, dump match.json.

• Play three full games yourself against yourself. Note every rule you had to look up. Those become UI copy.

Hour 40–48 · Ugly hotseat

• SVG circles + lines for the 7-node map. Click origin, click destination, type strength.

• Pass-and-play: show a “hand device to next player” interstitial so orders stay secret.

• Resolution log as a scrolling list. No animations yet.

• Stop. Play. Do not add factions, cards, or the world map until this 7-node game is actually tense.

07 — WEEK PLAN AFTER THE 48 HOURS

Week	Outcome that must exist
2	Cards + tactical move + elimination + victory on the knife-fight map. 10 logged games.
3	World-42 graph + original names. Income retune. First external playtest (3 humans).
4	Order UX that does not feel like filing taxes. Legal-move highlighting. Undo before lock.
5	Resolution cinema: arrows, vacated origins pulsing, spoils clearly called.
6	Bot v0: heuristic that values exposed origins and refuses suicidal vacate 70% of the time.
7+	Net lockstep or print-and-play PDF. Only if week 3 playtests say “again.”

08 — UI PRINCIPLES

• **The map is a status board, not a diorama.** Army counts must be readable at a glance. Vacated origins must look vacated.

• **Order sheet is a war plan.** List of committed attacks with origin remaining after commit shown in red if it drops to 1.

• **Lock is sacred.** After lock, no edits. Show a 3-second “sealed” state before resolve.

• **Resolution is the payoff.** Step through the five-stage queue. Let players watch a mass invasion collapse a capital. Autoplay with pause.

- **No hidden engine.** Every casualty line in the log should be reconstructible from the rules in section 03.
- **Color is for players, not decoration.** Colorblind-safe patterns on army tokens.

09 — WHAT TO TUNE VS WHAT TO LEAVE ALONE

Knob	Tune?	Notes
Attackers vacate origin	Never	Soul. If games feel harsh, teach better, don’t patch the rule.
Tie goes to attacker	Almost never	Gives SameTime its lean. Revisit only if defense is hopeless.
Dice / volley casualty	Yes	Primary balance lever.
Income divisor and cap	Yes	Second lever. Cap stops front-loading.
Region bonus values	Yes	Make the juicy region a trap if vacated.
Surge existence	No	Keep it. It is the reward for a clean breakthrough.
Surge timing (last)	No	Stops surge from invalidating other people’s primary plans mid-queue.
Card escalation	Yes, late	Only after income feels right.
Fog of war	Later	Not MVP. SameTime already has hidden intent.
Faction powers	Later	Will break balance if added before the vanilla game sings.

10 — TEST ORACLES (WRITE THESE FIRST)

If a future you “simplifies” resolve.ts and these still pass, you probably did not break the game. If they fail, you did.

T1 Vacate. Red origin 5 attacks Blue 4 with strength 4. Green attacks the same Red origin with 3. After lock, Red origin has 1. If Green wins that fight, Green owns the origin even if Red’s outbound packet later wins somewhere else.

T2 Mutual strike. Red 6 in A hits B, Blue 6 in B hits A. Clash resolves before either garrison is used as a defender of its own soil.

T3 Mass then spoils. Red 5 and Blue 5 both hit Green 3. Green dies. Red and Blue then fight over the empty land. Exactly one owner remains.

T4 Illegal leftover. Origin 3, two attacks of strength 2 each. Validator rejects. Origin must keep 1.

T5 Cap. Territory with 0 friendly neighbors cannot receive more than 1 reinforcement. Income beyond cap vanishes.

T6 Determinism. Same world snapshot + same sealed orders + same seed ⇒ identical world hash and identical log.

T7 Surge last. A surge target that is also someone’s primary destination is fought as a primary first. Surge hits whatever owner exists after Step D.

11 — BOT v0 (ONLY AFTER HOTSEAT WORKS)

Do not train a model. Write a greedy planner with three scored fears: (1) value of capturing a region choke, (2) cost of leaving an origin at 1 adjacent to an enemy stack, (3) expected volley value of each legal packet size. Sample a few candidate order sets, pick the best. A bot that never vacates is too timid. A bot that always vacates is a piñata. The interesting bot vacates on purpose.

12 — PHYSICAL PORT (DEFERRED)

When the digital numbers stabilize, the board version is: order pads, one shared d6 set, a resolution-order reference card, and a first-player-token that is only used to break simultaneous bookkeeping ties (who reads the log aloud), never to take a full sequential turn. Do not design custom dice until you know combat is finished. Paper is a follower of the engine, not a co-author.

13 — OPEN QUESTIONS (DO NOT BLOCK MVP)

- Is 38/42 the right domination threshold, or is a scored 8-round mandate better as default?
- Should region bonuses be paid only if the region's perimeter is garrisoned above a minimum?
- One tactical move is correct — but should sea lanes (dotted adjacencies) cost a card to use in the tactical phase?
- Do we ever show ghost arrows of last round's attacks as public history? Probably yes, as a teaching tool.
- Solo campaign vs skirmish bot — later. Skirmish bot first.

14 — DEFINITION OF DONE FOR v0.1 PLAYABLE

A build is "playable" when all of the following are true:

- Three humans can finish a knife-fight match without asking you how resolution works more than twice.
- The vacate-and-punish moment happens at least once and everyone groans for the right reason.
- A match.json exists for that game and replays to the same end state.
- No Math.random in packages/engine.
- README can start the hotseat client in one command.
- You still want a second game after the first. If you don't, the numbers are wrong or the UX hid the drama. Fix those before adding content.

Standing order: engine first, drama visible, vacate stays lethal. Everything else is decoration.

End of v0.1 bible. Amend only with a dated changelog and a failing test that justified the change.